

[Figure 9 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 2** (the two-layer target from Fig. 6b, nominal SM values $\mu^0 \approx 3$ MPa, $\mu^1 \approx 10$ MPa), using 3, 6, and 9 surface displacement datasets as inputs.

Explicit (left column): bilayer structure recovered with a clean interface. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 11.49]$ MPa; (b) 6 fields, SM $\in [1.00, 11.63]$ MPa; (c) 9 fields, SM $\in [1.00, 11.47]$ MPa.

Implicit (right column): bilayer structure recovered but with a visibly rougher interface between the two layers; more artifacts at the top edge. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 10.95]$ MPa; (e) 6 fields, SM $\in [1.00, 10.77]$ MPa; (f) 9 fields, SM $\in [1.00, 11.13]$ MPa.

Takeaway: both methods recover the two-layer structure; the explicit method produces a cleaner interface, while the implicit method's interface has more small-scale roughness. Slight overshoot in the top-layer SM values (colorbar max above the nominal 10 MPa) is visible in all six panels.

Fig. 9. The reconstructed results with varying 3, 6 and 9 surface displacement datasets corresponding to Sample 2 (Without noise, Unit: MPa).

[Figure 10 — placeholder]

Six-panel side-by-side comparison of the **Explicit Inverse Method** (left column) vs. **Implicit Inverse Method** (right column) on **Sample 3** (the three-layer target from Fig. 6c: blue bottom layer $\mu^0 \approx 3$ MPa, green middle layer $\mu^1 \approx 6$ MPa, red top layer $\mu^2 \approx 10$ MPa), using 3, 6, and 9 surface displacement datasets as inputs.

Explicit (left column): three-layer stacked structure recovered, with the middle (green) layer visibly distinct from the top (red) and bottom (blue) layers. Colorbar ranges: (a) 3 fields, SM $\in [1.00, 10.49]$ MPa; (b) 6 fields, SM $\in [1.00, 10.35]$ MPa; (c) 9 fields, SM $\in [1.00, 10.44]$ MPa.

Implicit (right column): three-layer structure recovered but with fuzzier interfaces between adjacent layers. Colorbar ranges: (d) 3 fields, SM $\in [1.00, 10.11]$ MPa; (e) 6 fields, SM $\in [1.00, 10.26]$ MPa; (f) 9 fields, SM $\in [1.00, 10.19]$ MPa.

Takeaway: as layer count increases ($1 \rightarrow 2 \rightarrow 3$), the quality gap between explicit and implicit widens. Sample 3's three-layer structure is harder for the implicit method to resolve cleanly than Sample 2's two-layer structure. More displacement datasets ($3 \rightarrow 6 \rightarrow 9$) improve both methods but do not close the gap.

Fig. 10. The reconstructed results with 3, 6 and 9 surface displacement datasets corresponding to Sample 3 (Without noise, Unit: MPa).

slightly larger than that for the explicit inverse method (see Table 1). Furthermore, it's worth noting that the quality of the reconstruction achieved by both inverse methods improved with an increase in the total number of measurements. By utilizing 9 boundary displacement measurements, the average relative error could be reduced to less than 10 %. However, it should be mentioned that each layer was not accurately mapped as the homogeneous case when using the implicit inverse method.

Additionally, we increased the noise level to 5 % and observed that the reconstructed results obtained by the explicit inverse method still outperform those obtained by the implicit inverse method (see Fig. 14, Fig. 15, Fig. 16). For the explicit inverse method, the resulting relative error in the case with 5 % noise is generally larger than that in the case with 1 % noise (see Table 1). However, even with a 5 % noise level, the relative error for the explicit inverse method remains lower than that for the implicit inverse method